

Aim:-Prepare the connection of ceiling fan along with the regulator and vary the speed.

APPARATUS Required:-

S.No.	Name	Range/Rating	Type	Quantity
1	Fan motor (stator/rotor)	60W,0.6A,350rpm,230±10 %volts	1phase squirrel cage induction motor	1
2	Capacitor	2.5MFD±5%,440V AC 50Hz,MAX. temp.85	+ - (-)	1
3	Fan assembly (Canopy/suspension rob, blades etc.)	-	-	1
4	Fan regulator	(0 -350)rpm	Resistance/electronics	1/1
5	Test lamp	200W		1
6	Different tools	-	-	As per requirement
7	Multi-meter	Analog/digital	-	1
8	Voltmeter	(0 -300)volts	Moving coil	1
9	Ammeter	(0 – 1) Amp.	Moving coil	1
10	Connecting leads	-	-	As per requirement

THEORY:-

Ceiling Fan:-

A ceiling fan is a propeller having two or more blades, directly driven by an electric motor and intended for use with free inlet & outlet. It is provided with a device for suspension from Ceiling of a room so that the blades rotate in a plane to give uniform air circulation in the room.

According to the electric motor used, ceiling fans can be classified as follows:

1. DC FANS: DC fans use series motors and are generally used where DC supply is easily available as in trains, buses etc.
2. AC FANS: AC fans are most commonly used domestic devices, which are generally known. AC fans use single phase squirrel cage induction motor.

Different Parts of Ceiling Fan

Fig (1) shows the construction diagram of ceiling fan. It contains the following parts-

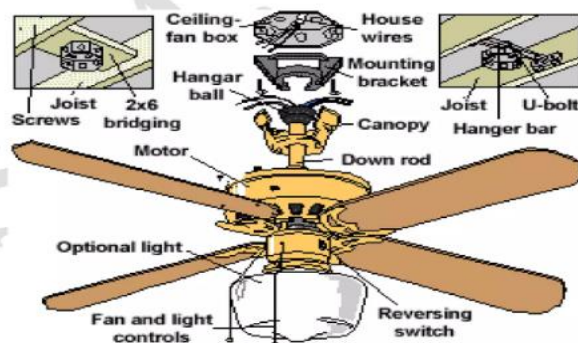


Figure:- Parts of Ceiling fan

Motor: The motor employed for ceiling fan is generally a single phase, permanent capacitor, split phase and squirrel cage induction motor. The stator consists of two windings

(a) Starting windings

(b) Running windings

Capacitor: A paper capacitor is provided at the top of the body of the fan. The value of capacitor depends upon the rating of the fan. The capacity of rating varies from 2.5 MFD to 3.0 MFD.

Blades: The ceiling fan is fitted with 3 or 4 angular blades of steel sheets or aluminum sheets. Aluminum blades are lighter and add efficiency of fan. The blades are fixed on the body of fan with the help of two machine screws.

Canopy: It is the shape of cup which covers the hook and capacitor. It is held on the suspension rod of the fan where it is screwed. Two canopies are used in each fan and are made of plastic.

A) One for covering hook.

B) Another for covering the capacitor.

Ball-Bearings: Ball-bearings are used for ensuring free movements of rotating parts. They are provided between rotating and stationary parts. The fans may have single or double ball-bearing. The double ball bearing is therefore a little costlier. The bearings are made of high quality steel and are regularly greased so as to provide long life.

Speed Regulator: It is used to control the speed of the fan. They can be either electrical or electronic type.

In electrical type regulator type speed regulator, speed is controlled by inserting and removing the resistance in series of fan.

In electronic type regulator, a triac is used to control the speed by varying input voltage. It is used for triggering.

WORKING PRINCIPAL

AC ceiling fan has single phase induction motor, which comprises two distributed windings: stator and a rotor (squirrel cage) when current is given to the motor, the magnetic field is experienced as a force in the rotor to move it right angle to the field at the blades attached with the rotor displace the air.

Circuit Diagram:-

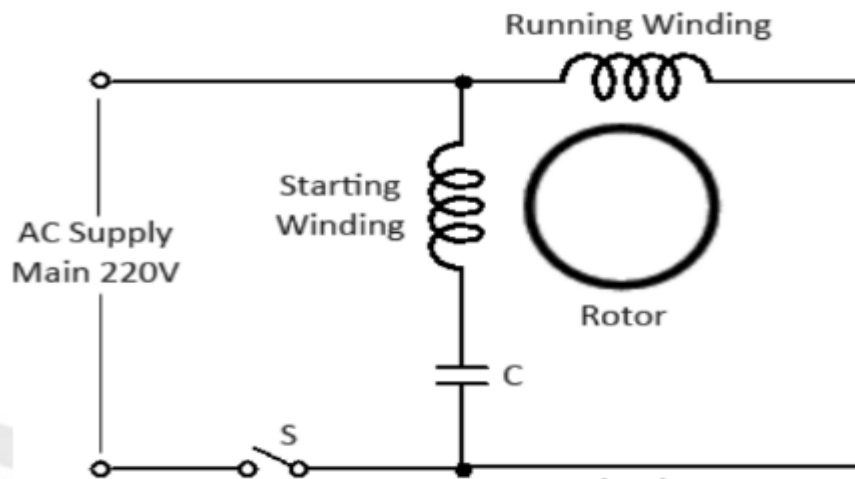


Fig-3.2: Ceiling fan capacitor connection diagram

Precaution:-

1. Connection should be proper. Loose connection is not allowed.
2. Do not touch any live wire or part.
3. Before switch on the supply, the connection should be checked by lab assistant.
4. Do not damage the terminals of part or wire etc.

Procedure:-

A) For testing-

1. Connect different circuit.
2. Check condition obtain for each and record.

B) For speed-

1. Connect ceiling fan as shown fig.
2. By varying knob of regulator, check the speed by visualization.

OBSERVATION TABLE

S.No.	Regulator resister	Speed(rpm)
1.		
2.		
3.		

Result:- We have successfully prepare the connection of ceiling fan with construction and working principal. Thus we have got the speed regulation at various regulator resister.

Viva voice Question:-

Que .1) Name the motor of ceiling fan?

Que.2) What is the function of capacitor in ceiling fan?

Que .3) What is name of insulation of ceiling fan?

Que. 4) Winding are fitted on stator or rotor of ceiling fan ?

Que.5) What happens if capacitor terminal interchanged?

Que.6) Why we use Capacitor in Ceiling Fan?